Autonomous Rover Terrain-Risk Classification

CS3006: Introduction to Data Science for Engineers

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Problem Statement & Objectives

The Challenge: Autonomous rovers must adapt their speed and motor torque based on the surface they are driving on to prevent stalling or mechanical damage.

Navigation-Risk Tiers:

- **Firm (Tier 0):** Concrete, tiles, wood. *Action: Full speed.*
- **Compliant (Tier 1):** Carpet, soft PVC. Risk of wheel sinkage and slip. *Action: Raise torque, slow down.*
- **Irregular (Tier 2):** Tile gaps. Risk of jolts and traction loss. *Action: Slow down, damp.*

Objective: Classify a 128-sample IMU window into one of these three tiers using a purely sensor-driven approach.

Dataset & Modality

Kaggle CareerCon 2019 (Help Navigate Robots)

- **Volume:** 3,810 windows $\times$ 128 sequential steps $\times$ 10 IMU channels.
- **Sessions:** 73 distinct recording sessions across 9 indoor floor surfaces.

Data Cleaning & Audit:

- Down-casted to `float32` for memory efficiency.
- Extreme outliers (robust $z > 6$) were **kept**, as impacts on tile gaps are genuine physical signals, not noise. Handled via train-fitted winsorization.

Feature Engineering: A Physics-Informed Approach

Instead of feeding raw sensors blindly, we engineered 149 features based on vehicle dynamics:

1. **Dropping Absolute Heading:** Raw quaternions encode which direction the rover faced (a session fingerprint). We extracted only the gravity projection (tilt).
2. **Vector Separation:** We split linear acceleration into:
   - *Vertical:* Captures bumps and vibrations.
   - *Horizontal:* Captures traction, drag, and slip.
3. **Spike Isolation:** Used a centered rolling mean (window=5) to isolate high-frequency impacts from the rover's slow, intentional movements.

Methodology: Preventing Data Leakage

The Flaw in Random Splitting:

- Shuffling 128-sample windows randomly lets the model memorize the unique background noise of a specific drive (a "session").
- **Evidence:** Our test script proved random folds artificially inflated Macro-F1 from 0.65 to 0.87.

Our Protocol: Stratified Group K-Fold

- Split strictly by `group_id` (Recording Session).
- **Split Ratio:** 60% Train (43 sessions), 20% Validation, 20% Test (15 unseen sessions).
- Cross-validation for hyperparameter tuning was conducted only on the training sessions.

Models & Architectures

We evaluated three models, increasing in complexity:

1. Multinomial Logistic Regression (Baseline)

Applied to the 149 tabular features with L2 regularization. Fast and highly interpretable.

2. Random Forest Classifier

Handles non-linear relationships and heavy-tailed spike distributions without scaling.

3. PyTorch 1D-CNN

Trained directly on the 11×128 raw/derived signal sequences.

- $3 \times$ (Conv1d $\rightarrow$ BatchNorm $\rightarrow$ ReLU) $\rightarrow$ Global Avg/Max Pooling.
- Restricted to **7,299 parameters** to prevent overfitting.

Risk-Aware Evaluation Metrics

Why Accuracy is a Trap: 60% of the training data is "Firm" terrain. A broken model that always guesses "Firm" gets 60% accuracy but stalls the rover on every hazard.

Our Primary Metrics:

- **Hazard Recall (Safety):** Of all actual hazards, how many did we catch? (False negatives = expensive mechanical stalls).
- **Hazard Precision (Efficiency):** How often were we right when we predicted a hazard? (False positives = harmless but inefficient braking).
- **Macro-F1:** Evaluates balanced performance across the severely imbalanced tiers.

Results & Model Comparison

Performance on the 15 Unseen Test Sessions:

Model	Macro-F1	Hazard Recall	Hazard Precision
Majority Baseline	0.250	0.000	0.000
Logistic Regression	0.452	0.704	0.576
Random Forest	0.478	0.734	**0.642**
PyTorch 1D-CNN	**0.473**	**0.781**	0.630

Note: The CNN effectively catches 78.1% of terrain hazards, outperforming classical models in safe navigation.

Conclusion Backed by Evidence: Why the 32% Error?

We conducted a deep dive into the misclassification rate of our best model:

- **Concentrated Errors:** 58% of all errors came from just 3 specific recording sessions.
- **The "Wood" Floor Problem:** 88 out of 92 errors on wood floors were incorrectly predicted as "Compliant" (soft).
- **Physical Diagnosis:** Wood floors in this dataset had a median peak rotation rate of 0.185 rad/s, which closely mirrors the 0.165 rad/s of soft carpets.
- **Conclusion:** An IMU measures vibration, not material stiffness. A smooth firm floor feels identical to a smooth soft floor. To improve beyond 78% recall, future rovers must fuse IMU data with wheel odometry (slip detection) and depth cameras.

Thank You

Questions?